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# Multidimensional analysis of climate-induced streamflow variability using CMIP6 data and advanced modeling techniques

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## ABSTRACT

The climate driven streamflow changes in Thailand's TC-Basin are assessed with CMIP6 GCMs, machine learning (Random Forest, XGBoost) and Bayesian Model Averaging (BMA). For precipitation ( $R^2 = 0.90$ , RMSE = 2.26–2.37 mm) and temperature ( $R^2 = 0.82$ –0.91, RMSE = 0.78–1.43 °C), RF/XGBoost achieved high accuracy. For precipitation downscaling, GFDL-ESM4 was suitable, and CNRM-CM6-1 was best for temperature. The BMA ensembles improved reliability (precipitation:  $R^2 = 0.84$ , RMSE = 4.24–5.45 mm; temperature:  $R^2 = 0.91$ , RMSE = 0.78–1.17 °C). Annual precipitation increased to 96,151.50 mm under SSP585, and maximum temperatures were 33.66 °C. Wet season (June–October) increases by 9–10% (higher flood risks), dry season (March–April) decreases by 40–43% (water scarcity), and cool, dry season flows increase by 33–56% are projected. Variation of wind speed was moderate ( $R^2 = 0.88$ , RMSE = 1.80–3.03 m/s), and humidity was stable ( $R^2 > 0.89$ ). This framework combines ML, GCMs, and BMA to quantify hydroclimatic shifts under climate extremes.

## ARTICLE HISTORY

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## KEYWORDS

Climate change; streamflow prediction; machine learning downscaling; Bayesian ensemble modeling

## 1. Introduction

Anthropogenic climate change has impacted global rainfall patterns, resulting in changes in hydrological cycles and weather systems (Wang et al. 2024) while increasing the frequency and intensity of extreme hydrological events, including droughts and floods (Kundzewicz et al. 2018; Fu et al. 2023; Xiong and Yang 2024). Between 1960 and 2019, about 17% of Earth's land surface changed, including losing 0.8 million km<sup>2</sup> of forests and expanding agricultural areas by 1.0 million km<sup>2</sup> (Winkler et al. 2021). These land

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use/land cover (LULC) shifts, primarily driven by human activities, affect watershed hydrological systems in various ways (Koneti et al. 2018). Understanding the spatiotemporal patterns of LULC changes is essential for addressing climate change and guiding related discussions (Change 2019). However, challenges persist in analyzing global LULC trends due to data limitations and the inherent uncertainties in transformation records (Prestele et al. 2017). These climate change-induced influences affect the water resource management and planning sectors (Waqas et al. 2021). Recent studies, such as assessing climate change impacts on the Kankai River Basin in Nepal, have shown significant increases in temperature and precipitation under future scenarios, resulting in a projected streamflow rise of 11.65–36.74% by the end of the century (Lamichhane et al. 2022). Therefore, effective water resource management is imperative in adopting the global water scarcity crisis, which has become increasingly evident over the past decade (Greve et al. 2018). A precise streamflow estimation facilitates strategically planning available water resources (Ngo et al. 2024). Long-term streamflow forecasts weekly, monthly, seasonal, and annual timescales—play a pivotal role in optimizing the operation of reservoirs, irrigation systems, and hydropower facilities (Liang et al. 2018).

Global Climate Models (GCMs) are key tools for projecting the long-term impacts of climate change by simulating physical processes over extended temporal horizons (Bader et al. 2008; Flato et al. 2014). These GCMs are employed to develop climate scenario projections from the Coupled Model Intercomparison Project Phase 6 (CMIP6) of the World Climate Research Program (WCRP). CMIP6 builds upon the insights of CMIP5 by integrating Representative Concentration Pathways (RCPs) with progressive socio-economic development trajectories (O'Neill et al. 2017). These shared socio-economic pathways (SSPs) offer distinct narratives of how global societies may evolve in the context of climate change and policy interventions (O'Neill et al. 2017). The latest generation of GCMs under CMIP6 provides a more detailed and accurate representation of Earth's physical systems than earlier models (Schlund et al. 2020). CMIP6 incorporates coordinated experiments to minimize biases, thereby enhancing the precision of its climate projections relative to prior iterations (Reddy et al. 2023). A study by Subedi et al. (2024) assessed future streamflow responses in the Tamor River Basin, Nepal, under CMIP6 climate projections using the SWAT model, revealing that increases in precipitation and temperature under SSP245 and SSP585 scenarios could evaluate annual streamflow by up to 20% by the end of the twenty first century (Subedi et al. 2024). However, the computational intensity required for large-scale simulations constrains these models to coarse spatial and temporal resolutions (Bader et al. 2008). This limitation poses challenges in addressing critical societal systems, which operate at regional and local scales where the resolution of GCM projections is insufficient (Kendon et al. 2021). Downscaling methods refine GCM outputs to bridge this gap, providing higher-resolution climate projections (Humphries, Waqas, Hlaing, Dechpichai, et al. 2024).

Statistical and dynamic downscaling methods to address the spatial scale disparity between the coarse-resolution outputs of GCMs and hydroclimatic variables at the catchment scale (Wilby and Wigley 1997). Statistical downscaling involves deriving empirical relationships that link GCM outputs with catchment-scale hydroclimatic data, effectively bridging the spatial resolution gap (Benestad et al. 2008; Wilby and Dawson 2013). In contrast, dynamic downscaling employs physics-based equations to achieve the same objective (Kim and Ivanov 2015). Statistical downscaling is widely used due to its computational efficiency and simplicity (Vrac et al. 2012; Waqas, Humphries, Chueasa, et al. 2024; Waqas, Humphries, Hlaing 2024), making it a more accessible alternative to dynamic downscaling. According to Wilby and Dawson (2004), statistical downscaling

methods can be categorized into three main groups: regression-based, weather classification, and weather generator-based (Wilby and Dawson 2004). Regression-based methods have gained significant attention due to their ease of application and straightforward implementation (Khan et al. 2024).

Regression-based methods, including ML models such as artificial neural networks (ANN), recurrent neural networks (RNN), Regression Trees, AdaBoost, Gradient Boosting, Decision Trees (DTs), and Random Forests (RFs), are employed in statistical downscaling due to their relatively low data demands (Chen et al. 2010; Liu et al. 2016; Pang et al. 2017; Nourani et al. 2019; Hosseini Baghanam et al. 2024). Recent studies, including Khan et al. (2024), have evaluated various ML algorithms such as RNN, Regression Trees, AdaBoost, and Gradient Boosting under different SSPs. AdaBoost and RNN demonstrated exceptional predictive capabilities, achieving coefficients of determination surpassing 0.85 during testing phases (Ullah et al. 2023). Similarly, Singh et al. (2023) applied ML models, including RF and one-dimensional CNNs, to simulate streamflow in the Sutlej River Basin. Their research highlighted seasonal streamflow variations driven by precipitation and temperature changes, as projected by CMIP6 GCMs under SSP scenarios (Rangra et al. 2024). Addressing the coarse resolution of GCMs, these studies employed bias-corrected CMIP6 data with a high spatial resolution ( $0.25^\circ \times 0.25^\circ$ ) alongside ML-based statistical downscaling techniques, significantly improving regional hydrological predictions (Wing et al. 2024). eXtreme Gradient Boosting (XGBoost) is recognized for its ability to model complex, non-linear relationships between environmental factors and precipitation patterns (Tao et al. 2022). Due to its robustness and ease of use, XGBoost has been widely utilized for the downscaling of various hydrometeorological variables (Zhu et al. 2023), land surface temperature (Xu et al. 2021), and air temperature (Zhang et al. 2024). XGBoost has demonstrated superior performance in statistical downscaling tasks (Duan et al. 2023). In comparing four ML-based algorithms and three interpolation methods for precipitation downscaling, XGBoost achieved the highest accuracy among the models tested (Zhu 2024). Ali et al. (2023) also reported that XGBoost outperformed ANN in downscaling terrestrial water storage data derived from GRACE satellite observations (Ali et al. 2023). Furthermore, recent studies have applied advanced hybrid ML models, such as LSTM-ALO, LSTM-INFO, and RVM-IMRFO, which have shown strong performance in hydrological time series forecasting (Adnan et al. 2019; Ikram et al. 2023). These models have gained preference in downscaling, and this study employed RF and XGBoost due to their lower input data requirements, which is a critical advantage in scenarios where data scarcity is a significant limitation.

Furthermore, according to Hlaing et al. (2024), the TC-Basin is ungauged. To this end, they used ML-based parameter regionalization to predict the main hydrological variables. The Soil and Water Assessment Tool (SWAT) was used to simulate streamflow in the basin with an accuracy of  $R^2 = 0.85$  and  $0.64$  during the testing and validation stages, respectively (Hlaing et al. 2024). Another study used the SWAT model to evaluate the upper TC-Basin's flow regime, water quality, and sediment-related problems (Yasin and Clemente 2014). Similarly, a study in the TC-Basin examined the changes in climate and land use on the water resources using SWAT modeling in the basin and their importance (Monprapussorn 2018). The physically based hydrological model, the SWAT, has been widely utilized for assessing basin hydrology under various climate change scenarios at global and regional levels (Kumar et al. 2017; Chun et al. 2018; Liang et al. 2018; Ait M'Barek et al. 2023; Ikram et al. 2023; Mariani et al. 2024; Zhao et al. 2024).

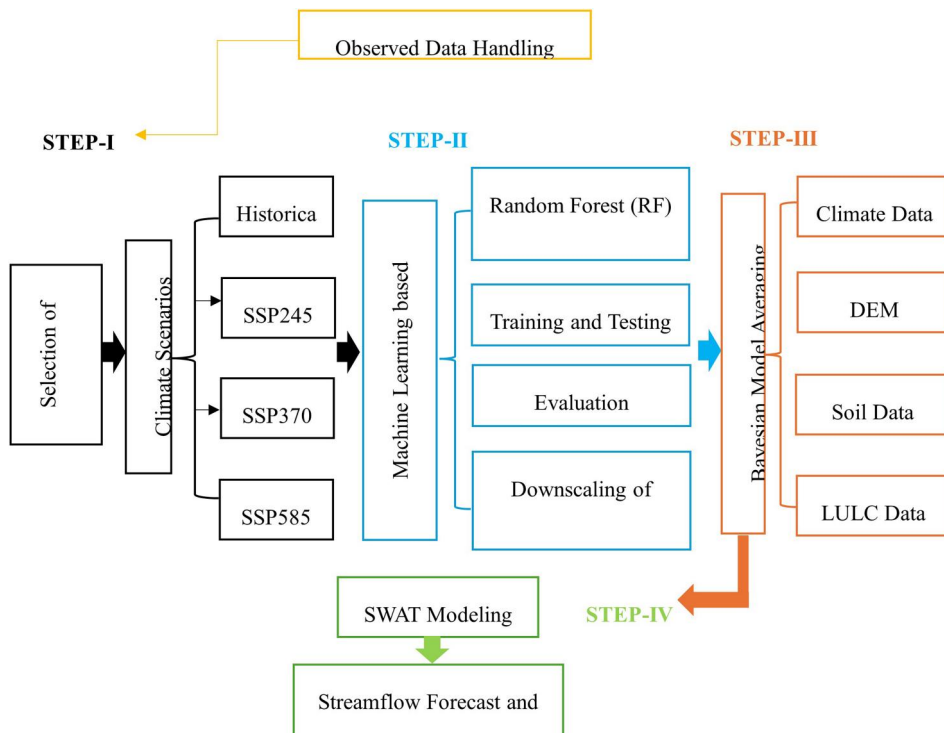
Despite significant advancements in hydrological modeling, uncertainties in GCM projections, data sparsity, and hydrological-climate interactions make it difficult to predict

streamflow under changing climatic circumstances. This work enhances streamflow estimates in the TC-Basin by merging a set of CMIP6 multi-model climate projections, BMA, and ML-based statistical downscaling. In contrast to previous research, this study provides a thorough analysis of future streamflow patterns by integrating a multifaceted assessment of the effects of LULC and climate change on streamflow. The study also aims to assess the influence of climate change on streamflow by utilizing CMIP6 GCMs alongside historical data. LULC was analyzed for 2012, 2017, and 2022 for the TC-Basin, with projections for potential future changes.

Additionally, the study evaluates future hydrological patterns under the combined effects of climate change and LULC modifications, addressing the reliability and uncertainty associated with these forecasts. The paper is organized as follows: [Section 2](#) provides an overview of the multidimensional framework, while [Section 3](#) describes the results from ML downscaling of climate data, BMA ensemble methods, and projections derived from CMIP6 GCMs for the TC-Basin. [Section 5](#) discusses uncertainties associated with hydrologic projections and land-use changes, and [Section 6](#) concludes with the key findings and recommendations for water resource management of TC-Basin.

## 2. Materials and methods

This study employs a multidimensional framework integrating ML and BMA modeling to evaluate streamflow variability under climate change scenarios ([Figure 1](#)). The methodology begins with selecting GCMs-CMIP6 datasets for historical and SSPs 245, 370, and 585. ML-based downscaling models, RF and XGB, refine coarse climate projections into high-resolution variables through training, testing, and evaluation processes. Downscaled



**Figure 1.** Overall methodology to assess the streamflow variability.

variables are then integrated with geospatial datasets, including DEM, soil characteristics, and LULC data, within a BMA framework to reduce uncertainties—the SWAT models and forecasts streamflow, enabling projections of future hydrological behavior.

2.1. Study area

The study focuses on the TC-Basin, a distributary of the Chao Phraya River, which lies within Thailand’s central plains, covering nearly 14,000 km<sup>2</sup> and featuring elevations from sea level to above 1,300 m (2004, May). The basin is home to about 2 million people and is indispensable for agriculture, transportation, and recreation (Hlaing et al. 2024). However, its ecosystems face degradation from urban and agricultural wastewater discharge. Originating in Chai Nat province, the river flows southward over 300 km, playing a crucial role in regional water resource management. It receives an annual average inflow of 1,350 million m<sup>3</sup> (Department 2018). The river basin’s average annual rainfall ranges between 1,200 mm and 1,600 mm, and temperature variations from 19 °C to 38 °C. Seasonal changes significantly impact river discharge, with peak flow occurring during the monsoon season (Hlaing et al. 2024). Figure 2 shows the geographical locations of the river basin, along with its LULC, DEM, and soil characteristics map.

2.2. Data collection

The datasets used in this study cover a range of daily meteorological (Precipitation, Temperature, Relative Humidity (R<sub>H</sub>), Wind Speed), hydrological, physical variables, and CMIP6-GCMs datasets, which are essential for accurate streamflow projections and prediction within the basin. Meteorological data, including precipitation, temperature, wind speed, and humidity, were collected daily from 1993 to 2022 and validated by the Thai Meteorological Department (TMD). Streamflow records for the gauged basins were

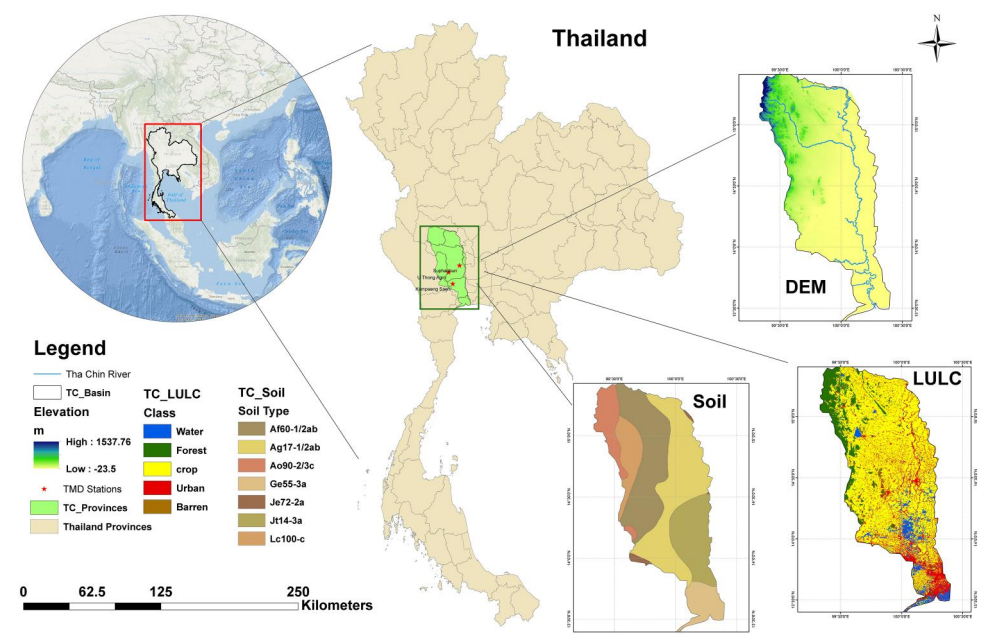


Figure 2. Geographical locations of the TC-Basin and its LULC, DEM, and soil characteristics map.



**Table 1.** Meteorological, hydrological, physical dataset, and sources.

| Data types                 | Data sources                                | Remark                                                      |
|----------------------------|---------------------------------------------|-------------------------------------------------------------|
| Meteorological Data        | Thai Meteorological Department (TMD)        | 1993–2022                                                   |
| DEM and Topographical data | Open Topography                             | 30-m resolution Copernicus DEM (European Space Agency 2021) |
| Soil Data                  | Digital Soil Map of the World (DSMW)        | 1:5,000,000 (FAO 1995)                                      |
| LULC Data                  | Sentinel-2 10 m Land Use/Land Cover         | 2017–2022 (Esri n.d.)                                       |
| Initial Streamflow Data    | Royal Irrigation Department (RID), Thailand | 1993–2022                                                   |

obtained from the Royal Irrigation Department (RID), while the ungauged TC-Basin's data were estimated using regionalization methods. Physical attributes were characterized using a 30-m-resolution Digital Elevation Model (DEM) from Copernicus Global and a 10-m-resolution land use map from ESRI Global. Soil property data were sourced from the Digital Soil Map of the World (DSMW) at a 1:50,000 scale (Table 1).

### 2.3. CMIP6-GCMs selection and data collection

For the selection, CMIP6-GCMs, with their improved features, are frequently used for projecting future climates under the SSPs, including SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5 (O'Neill et al. 2016). In this study, CMIP6-GCMs are selected based on their abilities in past implementation in Thailand (Eyring et al. 2016). High-resolution models like CESM2 and CNRM-CM6 are assessed for their ability to accurately simulate local climate phenomena in different regions of Thailand, such as precipitation patterns and extreme events (Humphries, Waqas, Hlaing, Wangwongchai, et al. 2024; Humphries, Waqas, Hlaing, Dechpichai, et al. 2024; Chatklang et al. 2024). The GFDL and EC-Earth3-Veg-LR models have been selected due to their demonstrated reliability in simulating historical climate data (De Silva et al. 2023; Pimonsree et al. 2023; Kuinkel et al. 2024b). These GCMs offer high confidence in their ability to project future climate scenarios (Held et al. 2019; Doeschl-Wilson et al. 2020). MIROC6 and MPI-ESM are recognized for their robust ability to replicate oceanic and atmospheric processes (Chaithong 2024; Chaithong et al. 2024; Kuinkel et al. 2024a). These CMIP6-GCMs are helpful in research areas such as Southeast Asia that require detailed simulations of climate dynamics (Mauritsen et al. 2019; Tatebe et al. 2019). CMIP6-GCMs outputs are obtained from global data repositories like the Earth System Grid Federation (ESGF), which provides access to key variables and scenarios for research. Commonly retrieved climate variables include temperature, precipitation, wind speed, and RH. Based on CMIP6-GCM evaluation studies and considerations of model resolution and their respective strengths, 10 CMIP6-GCMs are identified and presented in Table 2.

### 2.1. ML-based downscaling

This study developed two ML-based downscaling models, RF and XGBoost. The RF and XGBoost algorithms are ideal for large datasets with many features and non-linearity (Pang et al. 2017; Ali et al. 2023; Zhu et al. 2023). RF is less prone to overfitting due to less or no hyperparameter tuning (Waqas et al. 2021). AdaBoost utilizes the number of weak learners to build a strong model for prediction (Duan et al. 2023). RF and XGBoost were selected in this study due to their high accuracy, computational efficiency, and

**Table 2.** Selected CMIP6 GCMs for downscaling of climate data.

| CMIP6-GCMs       | Institute                                                                                                                                                                 | Resolution  | Reference                    |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------|
| BCC-CSM2-MR      | Beijing Climate Center, China Meteorological Administration                                                                                                               | 1.1° X 1.1° | (Wu et al. 2019)             |
| CAMS-CSM1-0      | Chinese Academy of Meteorological Sciences                                                                                                                                | 1.1° X 1.1° | (Eyring et al. 2016)         |
| CESM2            | National Center for Atmospheric Research (NCAR), USA                                                                                                                      | 1.3° X 0.9° | (Danabasoglu et al. 2020)    |
| CNRM-CM6-1       | National Centre for Meteorological Research, France                                                                                                                       | 1.4° X 1.4° | (Voldoire et al. 2019)       |
| CNRM-ESM2-1      | Various European institutions coordinated by the Royal Netherlands Meteorological Institute                                                                               | 0.7° X 0.7° | (Doeschl-Wilson et al. 2020) |
| EC-Earth3-Veg-LR |                                                                                                                                                                           |             |                              |
| GFDL-ESM4        | Geophysical Fluid Dynamics Laboratory, USA                                                                                                                                | 1.3° X 1°   | (Held et al. 2019)           |
| MIROC6           | Atmosphere and Ocean Research Institute (The University of Tokyo), National Institute for Environmental Studies, and Japan Agency for Marine-Earth Science and Technology | 1.4° X 1.4° | (Tatebe et al. 2019)         |
| MPI-ESM1-2-LR    | Max Planck Institute for Meteorology, Germany, and Meteorological Research Institute, Japan                                                                               | 0.9° X 0.9° | (Gutjahr 2019)               |
| MRI-ESM2-0       |                                                                                                                                                                           | 1.1° X 1.1° |                              |

strong ability to capture non-linear relationships in climate data (Waqas et al. 2023). Compared to deep learning models, RF and XGBoost require less computational power while providing comparable performance, making them more practical for regional-scale applications. Similar findings were reported by Shrestha et al. (2025), where RF and XGBoost achieved reliable results with limited data, demonstrating their suitability for hydrological studies.

**2.1.1. RF model**

The RF model is a non-parametrical ML algorithm that can be employed for both regression and classification analyses due to its ability to analyze data with numerous variables and multiple classes (Breiman 2001). RF has become popular because of its high predictive power and low hyperparameter tuning needs (Waqas et al. 2021). The model builds several parallel decision trees in different computational cores, improving scalability and computational speed (Buschjager and Morik 2018). In RF, two primary parameters are crucial to the algorithm: the forest’s number of trees and randomly selected features at each node to decide the best split (Chan and Paelinckx 2008; Probst et al. 2019). The CART algorithm is adopted for split selection, which helps the model identify complex patterns in the data (Loh 2002).

This study implemented the RF model, which was built with 200 trees, a maximum feature selection of ‘sqrt,’ and a maximum depth of 20. The minimum samples for split nodes were two, while the minimum samples at the leaves were one. Bootstrap sampling was allowed to achieve more accurate model performance.

Moreover, the model used 3-fold cross-validation for tuning the parameters and the training and testing data ratio of 7:3. Feature engineering included polynomial features and scaling in addition to temporal variables such as a month, day of the year, and, i.e. lagged minimum temperature (Tasmin). To make the predictions physically reasonable, output fixing was used to prevent negative output. This configuration allowed the RF model to reach high predictive accuracy and reasonable computation time, which is the advantage of Python 3’s flexibility and extensibility.



### 2.1.2. XGBoost model

XGBoost is an enhanced version of Gradient Boosting that is observed for its computational speed and capacity to generate accurate models by gradually improving DTs (Xi et al. 2018; Wade and Glynn 2020). The algorithm is designed to minimize a cost function that includes a loss function for model performance evaluation and a penalty term to reduce overfitting (Zhu et al. 2023). In addition, XGBoost applies a second-order Taylor expansion to more accurately approximate the loss function and improve the model's accuracy (Mushava and Murray 2022). In addition, cross-validation is incorporated into the boosting process to estimate the validation errors and prevent overfitting at each step (Wade and Glynn 2020; Phan et al. 2021).

In this study, the XGBoost model was trained using the sci-kit-learn library, and the hyperparameters were fine-tuned to get the best results. The model used 200 trees, the maximum tree depth was set to 6, and the learning rate was set to 0.1 to optimize the model's convergence speed while minimizing the loss. Furthermore, the model used sub-sampling at 0.8 and colsample by tree at 0.8 to minimize overfitting cases and guarantee stability across iterations.

The data was split into 70% training and 30% testing sets, and 3-fold cross-validation was used to tune the hyperparameters. Output clipping was also applied to prevent predictions beyond the physical realm. This careful configuration ensured that the XGBoost model had high predictive performance while maintaining reasonable computation time and showed that sci-kit-learn could be used to implement a wide range of ML techniques. These models were assessed using evaluation metrics such as  $R^2$  (Equation 2), MAE (Equation 3), and RMSE (Equation 3), which are explained in [Supplementary Files ANNEXURE-I](#). Nash-Sutcliffe Efficiency (NSE) and Mean Bias Error (MBE) were not applied in the downscaling evaluation, as these metrics are primarily designed for hydrological model performance assessment rather than for evaluating the statistical accuracy of climate variable downscaling (Table 3).

### 2.2. Bayesian ensemble modeling

This study used BMA (Chipman et al. 2006) to evaluate the results from ten climate models downscaled for TC-Basin. These historical data sets (1993–2014) were bias-corrected using ML downscaling techniques against observed station data. BMA was used to produce probabilistic forecasts of hydrometeorological variables by taking advantage of the performance of individual models and minimizing the errors and variability (Chipman et al. 2006). These ensemble projections were then incorporated into the SWAT model to evaluate the effects of climate change on streamflow in the TC-Basin. BMA, derived from

**Table 3.** Hyperparameters, optimal configurations, and shared features of RF and XGBoost models for climate data downscaling.

| Parameters                     | Grid search range        | RF optimal configuration | XGBoost optimal configuration |
|--------------------------------|--------------------------|--------------------------|-------------------------------|
| Number of trees (n_estimators) | [100, 200]               | 200                      | 200                           |
| Max features                   | ['auto', 'sqrt', 'log2'] | 'sqrt'                   | –                             |
| Max depth                      | [10, 20, None]           | 20                       | 6                             |
| Min samples to split           | [2, 5]                   | 2                        | –                             |
| Min samples at leaf            | [1, 2]                   | 1                        | –                             |
| Bootstrap                      | [True, False]            | True                     | –                             |
| Learning rate                  | [0.01, 0.1, 0.3]         | –                        | 0.1                           |
| Subsample                      | [0.8, 1.0]               | –                        | 0.8                           |
| Colsample by tree              | [0.8, 1.0]               | –                        | 0.8                           |
| Scoring Metric                 | $R^2$                    | $R^2$                    | $R^2$                         |

Bayesian posterior probability theory, incorporates model performance into the weighting process and offers accurate unified projections:

$$P(\theta|D) = \sum_{m=1}^M P(\theta|D, M_m)P(M_m|D) \quad (1)$$

Where  $P(\theta|D)$  is the posterior predictive distribution,  $P(\theta|D, M_m)$  is the model-specific posterior predictive distribution, and  $P(M_m|D)$  is the posterior model probability, reflecting the relative performance of model  $M_m$  based on the data (Hoeting et al. 1999; Raftery et al. 2005). In this, the response variable is observed climate data, while the predictors are the outputs of GCMs presented in Equation 2:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon \quad (2)$$

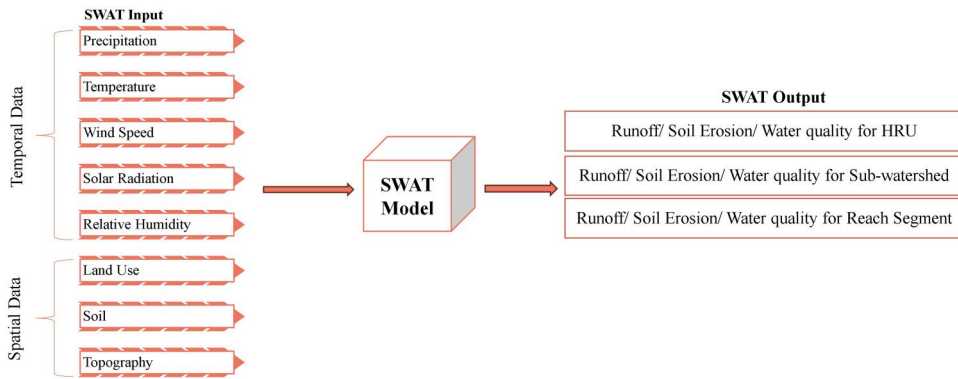
Where  $Y$  is the response variable,  $X_i$ ,  $i = 1, 2, \dots, k$  are the covariates,  $\alpha$ ,  $\beta_i$ ,  $i = 1, 2, \dots, k$  are the parameters of the model and the  $\epsilon$  is the random error (Irfan et al. 2024). BMA explicitly accounts for model uncertainty by allocating probabilistic weights that sum up to one, resulting in better-performing models on the ensemble forecast (Tebaldi and Knutti 2007). For the spatial distribution of climate variables, the Inverse Distance Weighting (IDW) method (Lu and Wong 2008) was used (ANNEXURE I in Supplementary Files).

### 2.3. Hydrological modeling with SWAT

SWAT is a process-based, semi-distributed hydrological model applied at various scales to predict stream flow, soil erosion, and water quality (Shope et al. 2014; Liang et al. 2018; Zhao et al. 2024). The application of SWAT follows a structured methodology comprising several critical stages: data pre-processing, model specification, estimation, testing, and result analysis. All these phases help improve the model's accuracy and reliability (Hlaing et al. 2024).

This study used SWAT to simulate streamflow in TC-Basin. A wide range of spatial and climatic input data was used to achieve high simulation accuracy. Climate data, including daily precipitation,  $T_{\min}$  and  $T_{\max}$ , wind speed, and RH, were obtained from the weather stations to simulate the hydrological processes such as streamflow (Hlaing et al. 2024), evapotranspiration (Chun et al. 2018), infiltration (Mariani et al. 2024), sediment transport (Feki et al. 2018), and surface runoff (Tibebe and Bewket 2011). A DEM was used for watershed delineation, stream network, and slope, which are crucial for flow direction and sediment transport (Datta et al. 2022). LULC data were used to capture spatial soil properties data, including texture, hydraulic conductivity, and water retention capacity, to simulate infiltration, percolation, and soil erosion processes (Ait M'Barek et al. 2023). The hydrological data used in this study include streamflow data from the gauging stations used in model calibration and validation. The hydrological parameter set applied in this study was derived from our previous research through regionalization methods (Hlaing et al. 2024), ensuring consistency and physical relevance for the TC-Basin (Figure 3).

Calibration was done using the Sequential Uncertainty Fitting (SUFI-2) method (Kumar et al. 2017), while statistical performance measures were NSE,  $R^2$ , and PBIAS. The sensitivity and uncertainty analyses improved the model by defining essential parameters and the degree of confidence in the predictions. The SWAT model produced useful outputs that included flow, erosion, and water quality, which were used to develop



**Figure 3.** Overview of SWAT hydrological modeling.

management policies for the watershed (Zhao et al. 2024). This approach is based on the conventional SWAT modeling practices to improve the study's validity and reliability.

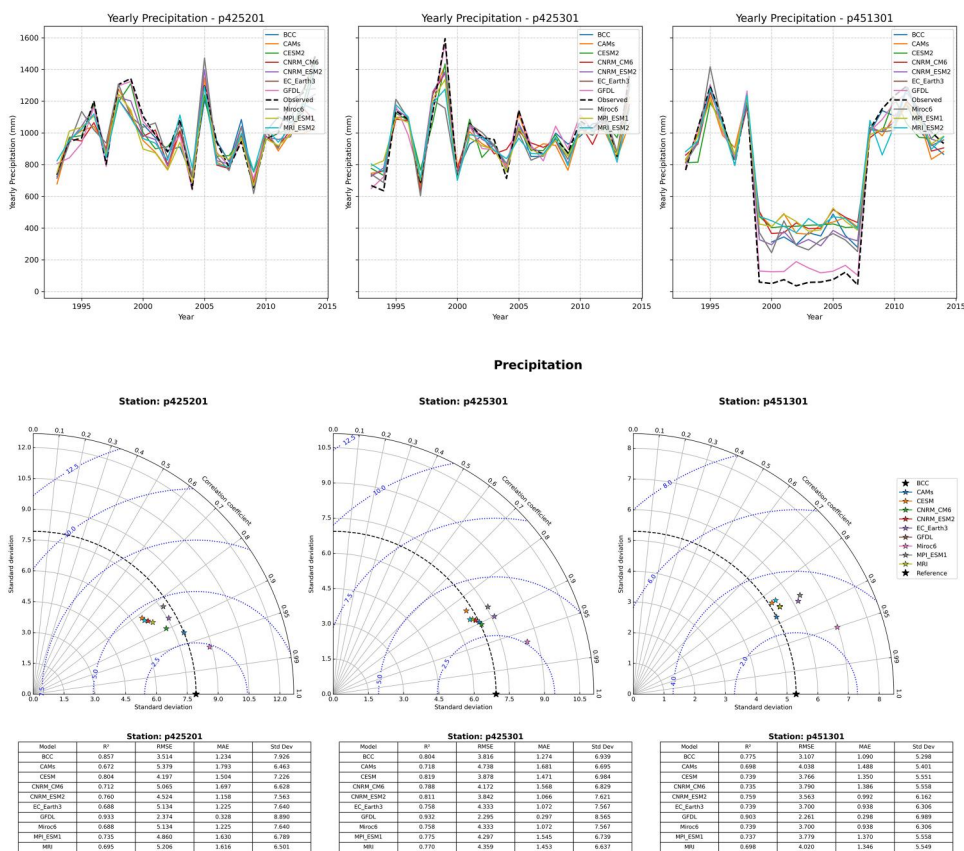
After downscaling the climate variables from ten CMIP6 GCMs RF and XGBoost, BMA was applied to create an ensemble of the downscaled outputs for each climate variable, reducing model uncertainty and improving predictive reliability. The BMA-ensembled climate data were then used as inputs to the SWAT model to simulate future streamflow under SSP245 and SSP585 scenarios. This approach ensures that the hydrological simulations reflect the combined strength of multiple climate models while minimizing individual model biases.

### 3. Results and discussion

#### 3.1. ML Downscaling of climate data

The results of the ML-based downscaling method RF and XGBoost Models for climate data demonstrate the nuanced strengths and limitations of various climate models in capturing localized climate variability. ML-based downscaling addressed this limitation by refining the large-scale model outputs to better represent regional climate features. Using statistical metrics such as  $R^2$ , RMSE, and MAE alongside visualization tools like Taylor diagrams (Humphries, Waqas, Hlaing, Dechpichai, et al. 2024). A robust comparative analysis has been conducted for key climate variables, including PCP, temperature ( $T_{\min}$  and  $T_{\max}$ ), wind speed, and  $R_H$ , across three stations in the TC-Basin: p425201 (Suphanburi), p425301 (U Thong Agro), and p451301 (Kampaeng Saen).

In downscaling, CMIP6-GCM, GFDL emerged as the most reliable GCM for PCP downscaling across all three stations (Figure 4a). It achieved  $R^2 = 0.90$  at all stations, indicating a strong correlation between the observed and predicted PCP. Moreover, the GFDL GCM exhibits the lowest error metrics, with RMSE ranging from 2.26 to 2.37 and MAE between 0.30 and 0.33. It indicates high precision in capturing PCP patterns with minimal deviations from observed data. The Taylor diagrams further reinforce the model's robustness, illustrating its strong alignment with the observed standard deviation and minimal centered RMSE differences. These findings highlight the GFDL model's capacity to capture complex PCP dynamics, making it a robust tool for TC-Basin. In contrast, other models, such as BCC, CAMS, and CESM2, demonstrate reasonable but comparatively weaker performance, as indicated by their lower ( $r$ ) and higher error metrics (Figure 4b).



**Figure 4.** (a) Observed and downscaled climate models for yearly precipitation; (b) evaluation of downscaled climate models for precipitation.

The CNRM-CM6 model stands out for its substantial predictive accuracy for  $T_{\max}$  simulations (Figures S1a and S2a). Across all three stations, it achieves  $R^2 = 0.82$ , with the lowest RMSE ranging from 1.16 to 1.43 and MAE between 0.83 and 1.00. These metrics underscore its ability to replicate observed temperature variations effectively. The Taylor diagrams validate this assessment, showing minimal deviations from observed data and a prominent level of agreement in variability representation. GFDL and EC-Earth3 models also perform reliably, achieving comparable  $R^2$ , though they exhibit slightly higher RMSE and MAE. This consistency across models highlights their suitability for temperature simulations, though CNRM-CM6 consistently maintains an edge in accuracy and precision.

For  $T_{\min}$  downscaling, the evaluation underscores the superiority of the GFDL model (Figures S1b and S2b). Its (r) exceeds 0.90 at some stations, and its error metrics remain the lowest among the evaluated models. For example, its RMSE and MAE are lower than those of other GCMs. It showed the exceptional capability to predict  $T_{\min}$  patterns. Other models, including CNRM-CM6, EC-Earth3, and MIROC6, also perform well but with notable variations across different stations. This variability suggests that while these models are competent, the GFDL model offers a more consistent and reliable performance. Accurate simulations of  $T_{\min}$  are critical for understanding extreme events and their implications for climate-sensitive sectors, further emphasizing the value of the GFDL model for detailed climate assessments.

The analysis of wind speed simulations reveals more variability among the models ((Figures S1c and S2c). The GFDL model consistently outperforms others, achieving a ( $r$ ) of 0.91 at station p425201 and exhibiting the lowest RMSE (e.g. 1.10 at p425201 and 1.65 at p425301) and MAE (e.g. 0.25 at p425201). In comparison, other models such as BCC, CNRM-CM6, and EC-Earth3 show moderate to low performance, with ( $r$ ) around 0.50 to 0.65 and significantly higher RMSE and MAE at stations p425301 and p451301. These findings underscore the limitations of some models in capturing wind speed dynamics accurately, which is critical for applications in meteorological forecasting and renewable energy planning. The disparity in performance also highlights the need for further calibration and improvement in models, particularly for stations with complex wind dynamics.

$R_H$  simulations exhibit strong performances from models like CNRM-CM6, CNRM-ESM2, and MPI-ESM1 (Figures S1d and S2d). These models achieve high  $R^2$  (approximately 0.87 at p425201 and 0.88 at p425301) and maintain low RMSE (e.g. 2.36 to 2.89) and MAE (around 1.05). These metrics reflect a robust ability to replicate observed humidity patterns. EC-Earth3 and MPI-ESM1 also perform commendably, achieving comparable  $R^2$  and error metrics. The Taylor diagrams support these findings by closely aligning with observed variability and minimal deviations. Accurate  $R_H$  simulations are critical for understanding atmospheric moisture content and its implications for ecosystems, agriculture, and water resource management. The reliable performance of these models underscores their utility in advancing climate research focused on atmospheric moisture. This comparative evaluation of different CMIP6-GCMs by ML models provided GCMs like GFDL and CNRM-CM6 that outperform GCMs across multiple variables for TC-Basin. These findings offer significant insights for selecting the most suitable downscaling models for specific climate variables and regions, enabling more accurate and actionable climate projections. The robust performance of specific models, particularly GFDL, underscores their potential as key tools in regional climate studies and their broader implications for climate adaptation and mitigation planning.

### 3.2. Bayesian model Averaging of downscaled climate models

BMA was employed to enhance the predictive performance of downscaled PCP CMIP6-GCMs outputs. By assigning posterior probability weights to individual models based on their historical performance, BMA provided probabilistic predictions that outperformed single-model outputs. At station p425201, the BMA ensemble achieved a robust  $R^2$  of 0.84, with an RMSE of 4.24 and an MAE of 1.54, reflecting a strong ( $r$ ) and reasonable precision (Figure 5). At station p425301, performance slightly declined, with an  $R^2$  of 0.82, an RMSE of 4.28, and an MAE of 1.56, indicating substantial pattern capture despite marginally higher errors. However, at station p451301, the ensemble recorded a reduced  $R^2$  of 0.64, with increased RMSE (5.45) and MAE (2.17). It showed challenges in modeling PCP dynamics in TC-Basin. These results suggest that while BMA effectively integrates diverse model outputs, its performance is sensitive to regional variability, necessitating further refinement and local model tuning to enhance predictive accuracy.

Furthermore, Figure 6 compares yearly PCP after BMA ensembling with observed PCP and the performance metrics for each station, respectively. Figure 7 presents projections of yearly average PCP under SSP245, SSP370, and SSP585 scenarios for 2021–2050, 2051–2080, and 2081–2100. The SSP245 scenario exhibits stable trends with moderate variability, while SSP370 and SSP585 display increased variability and higher PCP peaks in later periods. Among the stations, p451301 exhibits the highest sensitivity under SSP585,

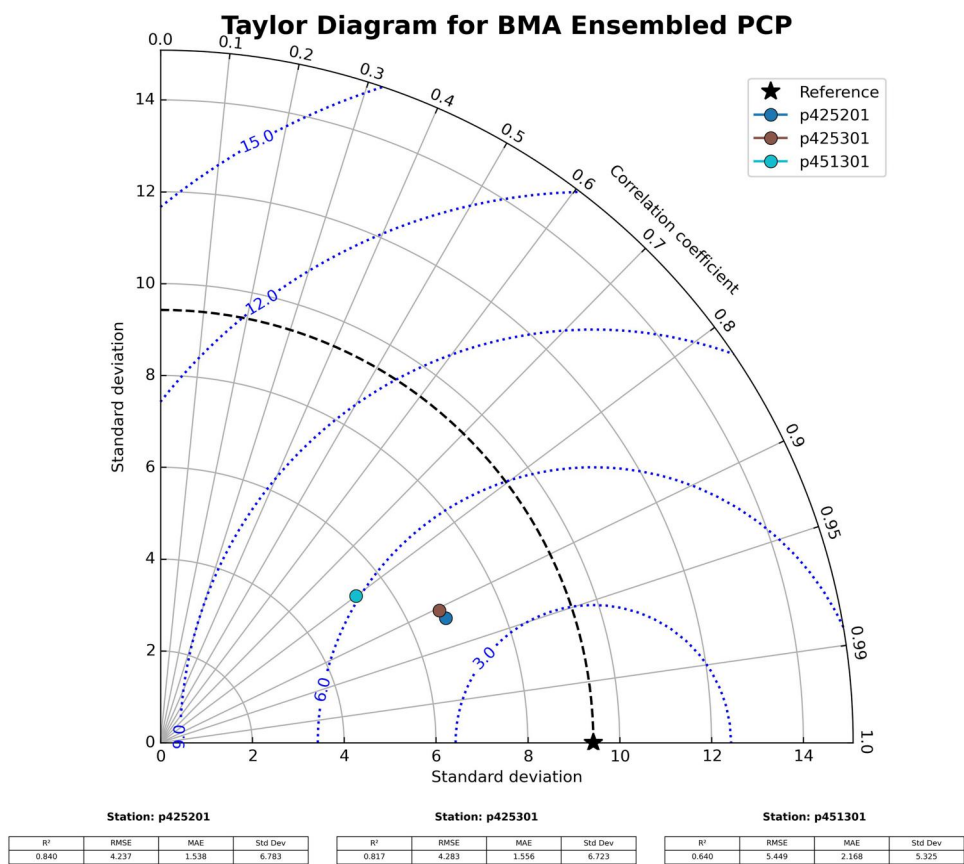


Figure 5. Performance evaluation of BMA ensembled precipitation.

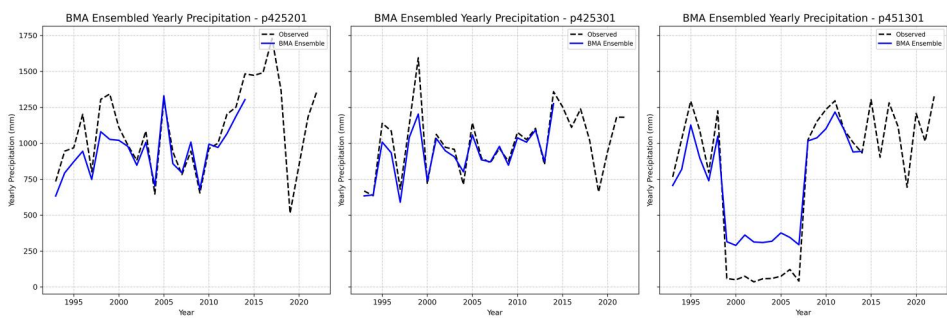
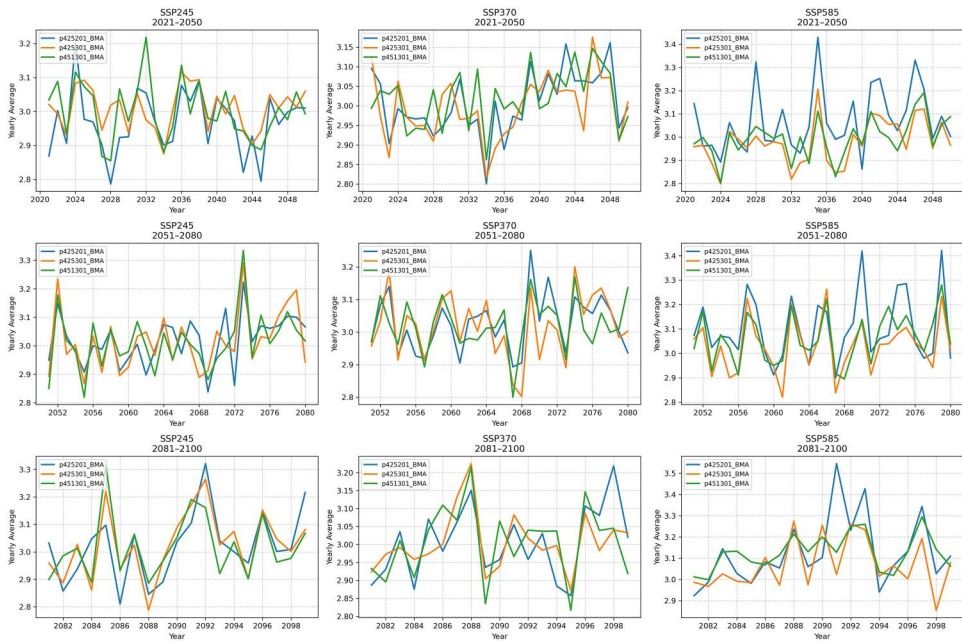


Figure 6. BMA Ensembled yearly precipitation.

underscoring its heightened vulnerability to extreme PCP events. These findings highlight the potential of BMA for improving ensemble-based PCP predictions in TC-Basin.

The BMA ensemble demonstrates commendable predictive accuracy in modeling  $T_{\max}$  (Figure S4). For station p425201, BMA attains an  $R^2$  of 0.85, supplemented by an RMSE of 1.08 and an MAE of 0.81, indicating robust model performance. At station p425301, the BMA exhibits its highest accuracy with an  $R^2$  of 0.87, an RMSE of 1.06, and an MAE of 0.78. This improvement suggests that localized factors at this station favor the predictive capabilities of BMA. Conversely, station p451301 reflects slightly diminished





**Figure 7.** BMA Ensembled yearly precipitation under SSP scenarios with 30years duration.

performance, with an  $R^2$  of 0.86 and increased error metrics of RMSE 1.35 and MAE 0.97. It shows more significant variability or complexity in local environmental conditions. These variations across stations underscore the BMA ensemble's ability to adapt to diverse datasets, though they also point to challenges in maintaining uniform accuracy under differing environmental influences. The comparison of yearly observed and BMA-enssembled yearly data is presented in [Figure S8a](#). Projections of yearly average  $T_{\max}$  under SSP245, SSP370, and SSP585 are presented in [Figure S3a](#) for three 30-year periods: 2021–2050, 2051–2080, and 2081–2100. The results reveal nuanced trends, with SSP245 exhibiting stable  $T_{\max}$  patterns, reflecting lower emissions and mitigation efforts.

In contrast, SSP370 and SSP585 scenarios display pronounced increases in temperature variability and elevated  $T_{\max}$  in later periods. Station p451301 demonstrates increased sensitivity under SSP585, with pronounced risks associated with extreme heat events. It suggests localized vulnerabilities to climate change. The progressive intensification of  $T_{\max}$  variability under high-emission scenarios underscores the imperative for preemptive measures to address the cascading impacts on ecosystems, agriculture, and water resources.

The performance of the BMA in predicting  $T_{\min}$  is equally notable, with high ( $r$ ) across all stations ([Figure S5](#)). For station p425201, BMA achieves an  $R^2$  of 0.91, an RMSE of 0.78, and an MAE of 0.56, indicating precise forecasts with minimal error margins. At station p425301, the  $R^2$  improves to 0.92, though the RMSE and MAE rise to 1.15 and 0.82, respectively, reflecting slightly reduced predictive precision. Station p451301, however, records a lower  $R^2$  of 0.81, with RMSE and MAE of 1.17 and 0.94, respectively, emphasizing some challenges in maintaining accuracy under varying local conditions. [Figure S8b](#) compares yearly observed and ensembled  $T_{\min}$  data and performance evaluations, affirming the ensemble's reliability in capturing  $T_{\min}$  variability. Projections of  $T_{\min}$  under SSP scenarios, as shown in [Figure S3b](#), reveal minor variability, even under higher emission scenarios such as SSP370 and SSP585. This stability suggests that  $T_{\min}$  is less susceptible

to direct emissions impacts than  $T_{\max}$ . Variations among the stations remain negligible, indicating uniform responses to changing climatic conditions across the study region.

The BMA performance in modeling wind speed is illustrated in (Figure S6). At station p425201, the model demonstrates strong predictive capacity, with an  $R^2$  of 0.88 and error metrics of RMSE 1.80 and MAE 1.33. At station p425301, the  $r$  decreases slightly, with an  $R^2$  of 0.79, while error metrics rise to RMSE 2.43 and MAE 1.98. Station p451301 presents the most significant challenge, with an  $R^2$  of 0.80 and the highest error metrics of RMSE 3.03 and MAE 2.44, indicating the influence of complex wind dynamics or localized environmental factors. Projections for wind speed under SSP scenarios Figure S3c highlight distinct spatial and temporal variability. SSP245 exhibits stable wind patterns, while SSP370 and SSP585 show increased variability, particularly in later periods. Station p451301 consistently records the highest wind speeds, suggesting a localized amplification of wind dynamics under higher emissions.

The ensemble's performance in predicting  $R_H$  is shown in (Figure S7). The BMA model achieves  $R^2$  about 0.89 across all stations, with error metrics stating its predictive precision. Station p425301 emerges as the best-performing site, with an  $R^2$  of 0.91 and the lowest RMSE and MAE, indicating favorable BMA conditions. Conversely, station p451301 reflects the highest error metrics, signaling the need for potential model refinements or deeper investigation into site-specific factors. Projected trends for  $R_H$  Figure S3d reveal minimal variation across scenarios, with SSP245 showing the most stability. SSP370 and SSP585 introduce slight increases in variability, though the overall trends remain consistent across stations. Station p451301 records marginally lower  $R_H$ , highlighting localized influences that may modulate atmospheric moisture responses. This uniformity suggests limited  $R_H$  sensitivity to emission pathways, offering some predictability amidst broader climate variability.

The BMA ensemble exhibits robust performance across temperature, wind speed, and  $R_H$  climate variables. BMA showed adaptability in integrating multiple model outputs. While localized complexities challenge uniform accuracy at some locations, the ensemble's overall reliability underscores its value in advancing climate modeling, informing mitigation, and shaping adaptation strategies.

### 3.3. Spatial analysis for BMA ensembled climate variables

The spatial analysis of PCP,  $T_{\max}$ ,  $T_{\min}$ , wind speed, and  $R_H$  across the TC-Basin reveals critical insights into the region's climate dynamics under different SSPs by employing Inverse Distance Weighting (IDW) in Equation 1. Figure 8 depicts PCP distributions divided into quartiles based on intensity, highlighting a stark contrast between historical and projected future conditions. Historical PCP, uniform across the basin, ranged from 15,611.90 to 20,848.90 mm annually. However, future scenarios under SSP585 predict an extreme intensification, with PCP up to 96,151.50 mm in the basin's western parts. This increase under high-emission scenarios underscores the likelihood of more frequent and extreme rainfall events, increasing flood risks, and changing hydrological patterns.

$T_{\max}$  projections across the TC-Basin analyzed using BMA demonstrate substantial spatial and temporal variability across historical conditions and future SSP scenarios (Figure 9). Historical  $T_{\max}$  ranged from 34.52 °C to 34.76 °C, concentrated in the central region of the TC-basin. Under SSP245, a moderate mitigation pathway,  $T_{\max}$  exhibits a slight decrease, ranging from 33.60 °C to 33.61 °C, indicative of modest cooling due to emission reductions. SSP370, representing less aggressive mitigation, shows a warming trend, with temperatures ranging from 33.60 °C to 33.61 °C. The most extreme scenario,

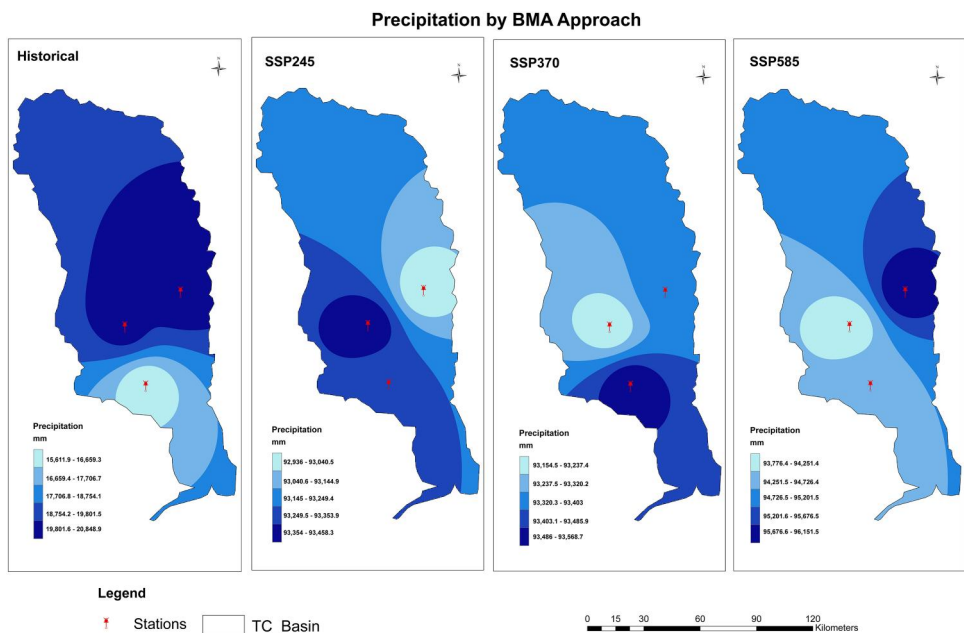


Figure 8. Spatial analysis of BMA ensemble precipitation.

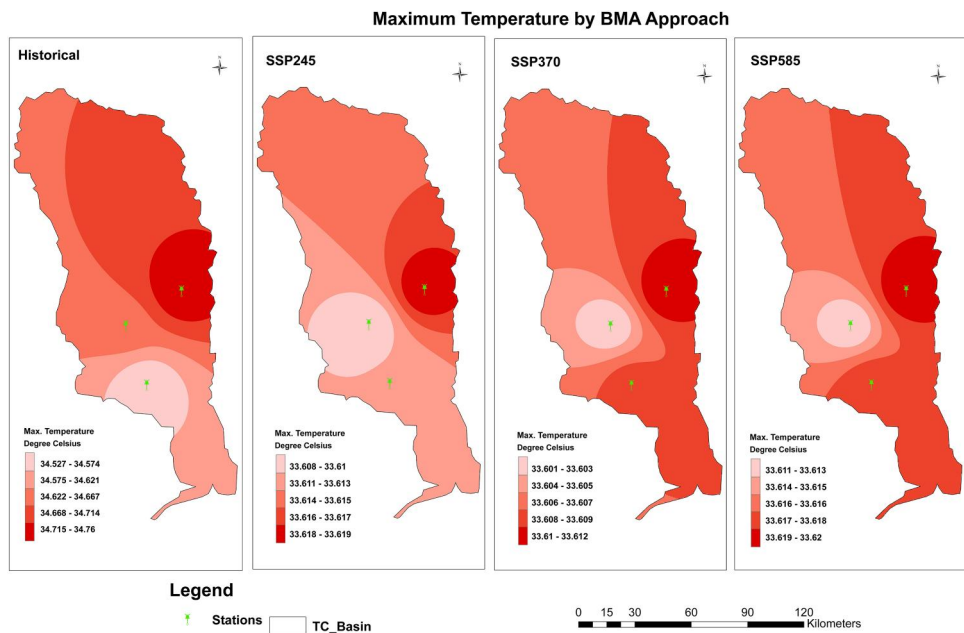
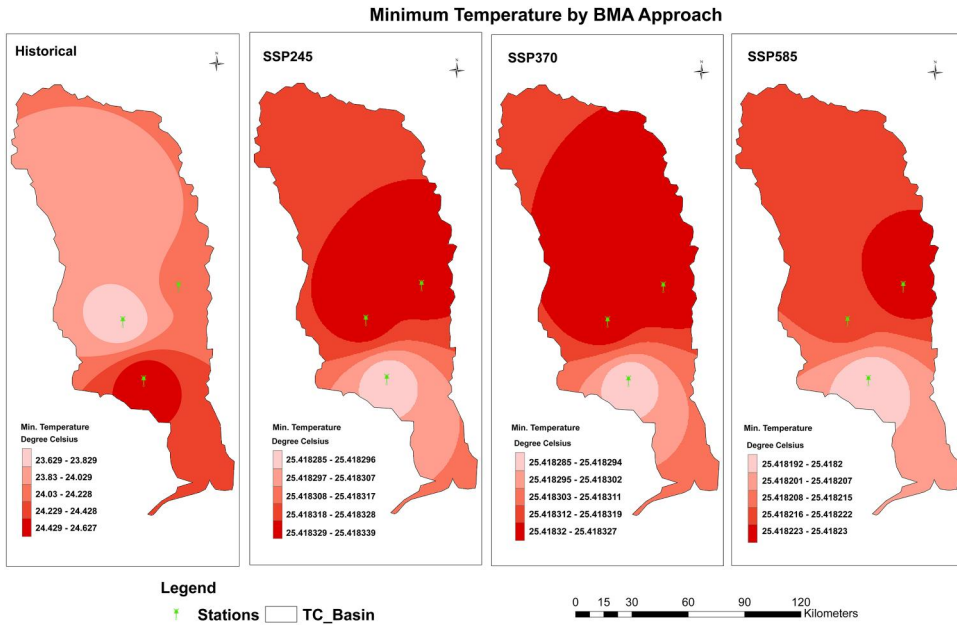


Figure 9. Spatial analysis of BMA ensemble maximum temperature.

SSP585, projects temperatures between 33.61 °C and 33.66 °C, indicating a significant rise in heat intensity in the northern and central regions of TC-Basin. This progression highlights the potential for increased heatwaves and extreme heat events, which could have profound implications for water resource management, agricultural productivity, and public health.



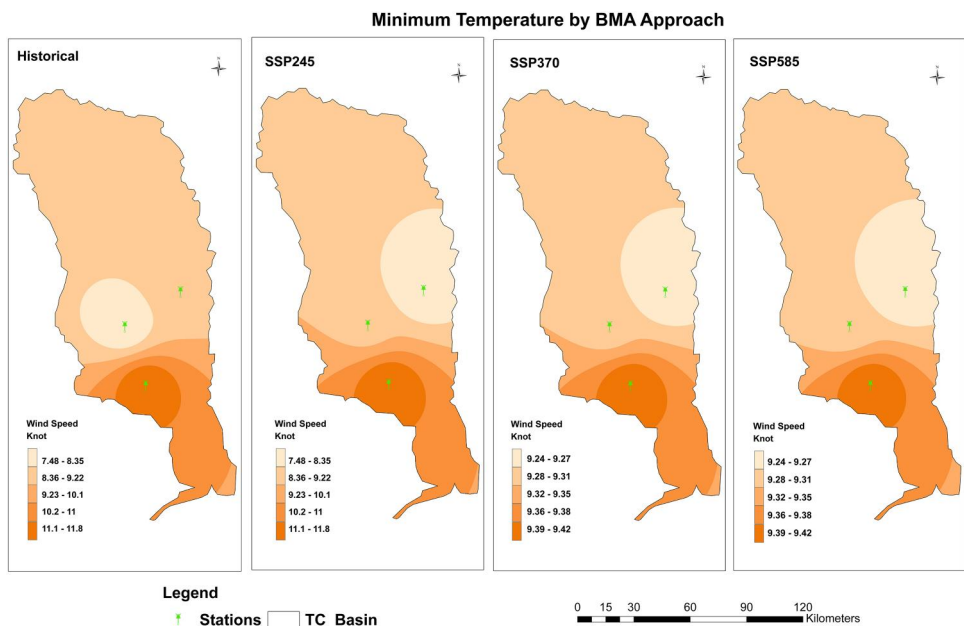
**Figure 10.** Spatial analysis of BMA ensembled  $T_{\min}$ .

Similarly, as depicted in [Figure 10](#),  $T_{\min}$  trends show a clear warming trajectory under future scenarios. Historically,  $T_{\min}$  ranged from 23.62 °C to 24.62 °C, with the coldest areas in the southern basin. Future projections under SSP245 indicate a modest increase, with temperatures rising from 25.41 °C to 25.41 °C, reflecting moderate mitigation effects. SSP370 presents further warming, with temperatures ranging from 25.41 °C to 25.41 °C, while SSP585, the most extreme scenario, forecasts a range of 25.41 °C to 25.41 °C. These projections suggest a trend toward reduced frost days and extended growing seasons. While beneficial for agriculture, it could intensify water demand and stress already strained water supplies.

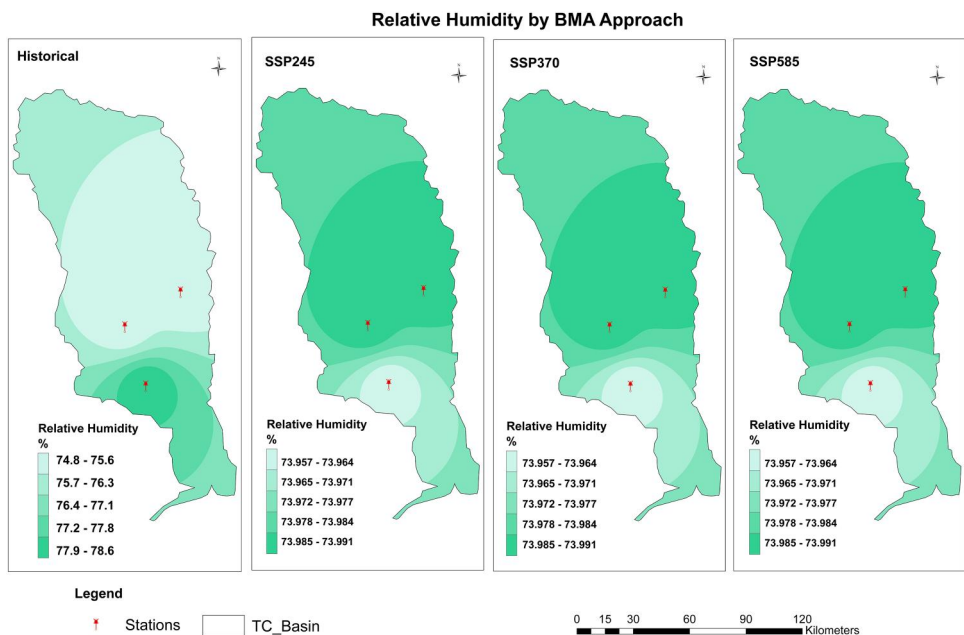
In [Figure 11](#),  $R_H$  projections exhibit a stable pattern across all SSP scenarios despite rising temperatures. Historical data indicates  $R_H$  levels from 74.8% to 78.6%, with the highest concentrations south of TC-Basin. Under SSP245, SSP370, and SSP585,  $R_H$  marginally decreases from 73.957% to 73.991%, implying drier conditions. This stability suggests a complex interaction between atmospheric warming and moisture retention, where rising temperatures do not change  $R_H$ . Consistent  $R_H$  levels underscore the necessity of considering various climatic variables in developing adaptive strategies to mitigate climate change impacts on the TC-Basin's hydrological cycles, agriculture, and ecosystems ([Figure 12](#)).

### 3.4. Impact of climate change on streamflow

Seasonal variability is highly sensitive to climate change in Thailand (Waqas, Humphries UW, Hlaing, et al. 2024). Seasonal patterns are illustrated in [Figure 13](#), with low flows during the dry season, rising to pre-monsoon months and peaking during monsoon season, then declining towards year-end. Projections based on SSP245, SSP370, and SSP585 show that monsoonal discharge will increase under SSP370 and SSP585 because of increased precipitation under high-emission conditions. The SSP585 flows to highlight the



**Figure 11.** Spatial analysis of BMA ensemble wind speed.

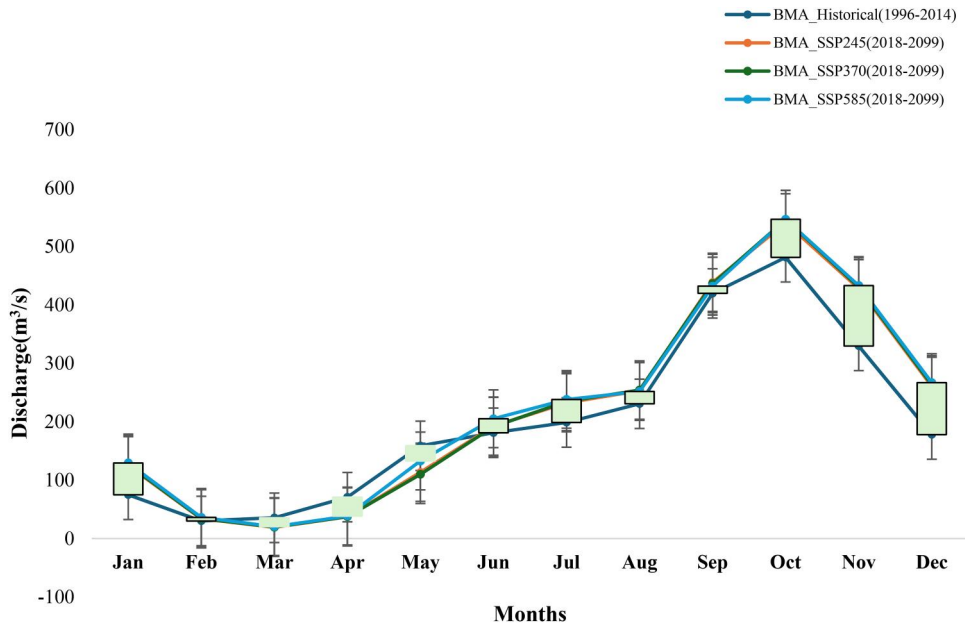


**Figure 12.** Spatial analysis of BMA ensemble relative humidity.

effect of high levels of greenhouse gases, with high uncertainty during the peak monsoon months because of the fluctuation in rainfall.

In the cool-dry season (November–February), streamflow reduces gradually, and the lowest streamflow occurs in February because of high temperatures and low rainfall (Figure S9a). Future projections are higher flows in November and December under





**Figure 13.** Historical and future monthly streamflow variability.

SSP370 and SSP585, possibly due to monsoon duration that can be used for irrigation. However, February is still very much a parched month, which only intensifies the water shortage problem. The hot-dry season is characterized by low flows in March and April due to high evapotranspiration and low precipitation, while the May flow is influenced by pre-monsoon rainfall (Figure S9b). Lower flows under SSP245 during May indicate delayed rainfall and higher evapotranspiration, contributing to water stress.

The highest flows are observed during the wet season, which lasts from June through October, with the most significant flow rate in September (Figure S9c). SSP370 and SSP585 increase monsoonal precipitation and streamflow, with the highest peak discharge resulting from SSP585. These conditions increase the likelihood of flooding, erosion, and sedimentation, affecting agriculture, infrastructure, and water quality. Small rises under SSP245 emphasize the advantages of low-carbon development. Future variability also suggests that future management of enhanced flows and the associated risks during the rainy season will require flexibility in the management strategies.

#### 4. Discussion: comparative analysis with related studies

This study employs a multidimensional approach integrating ML models and BMA to evaluate streamflow variability under climate change scenarios in the TC-Basin. Integrating RF and XGBoost algorithms for downscaling climate data presents a significant advancement compared to traditional methods, such as regression-based statistical downscaling (Wilby and Dawson 2004, 2013). While previous studies, including (Vrac et al. 2012), highlight the limitations of conventional downscaling techniques in capturing non-linear dynamics, this research effectively addresses these challenges by leveraging the robust feature selection and predictive capabilities of RF and XGB. Similar findings were reported by Zhu et al. (2023), where XGBoost outperformed other models in downscaling extreme precipitation events, underscoring the adaptability and precision of gradient-



boosting algorithms for complex climate datasets (Zhu et al. 2023). Compared to recent studies, such as the application of the optimally pruned extreme learning machine (OP-ELM) for daily streamflow prediction in the Fujiang River Basin, which demonstrated improved accuracy over models like ANFIS-PSO and MARS through kernel optimization and local data integration (Adnan et al. 2024), this study focused on long-term, scenario-based streamflow projections under climate change conditions using Random Forest, XGBoost, and Bayesian Model Averaging. Unlike short-term daily forecasting, our approach prioritizes capturing seasonal and interannual variability influenced by future climate scenarios.

Bayesian ensemble modeling in this study offers a robust method for integrating outputs from multiple CMIP6 GCMs. It aligns with Liu et al. (2016), who emphasized the superiority of ensemble methods like BMA in reducing uncertainty and enhancing the reliability of hydrological predictions (Liu et al. 2016). However, the present study advances the methodology by incorporating high-resolution CMIP6 outputs and integrating them with local geospatial datasets, a step that Liu and others identified as critical but underexplored. For example, this study's spatial variability in precipitation and temperature trends highlights localized impacts under different SSPs, which were less detailed in prior research using CMIP5 datasets (Kendon et al. 2021). The results of this research align with studies like Liang et al. (2018), which utilized the SWAT model for long-term streamflow forecasting (Liang et al. 2018). However, this study's integration of ML and BMA with SWAT represents a methodological leap, enhancing the model's capacity to simulate streamflow under varied climate scenarios. For example, incorporating LULC data and high-resolution DEMs ensures that the model captures fine-scale hydrological processes, addressing gaps noted by Koneti et al. (2018) regarding the impact of LULC changes on runoff dynamics (Koneti et al. 2018).

One key distinction lies in the study's emphasis on the predictive accuracy of CMIP6-based projections. While Reddy et al. (2023) evaluated CMIP6 models for streamflow simulations in the Godavari Basin, their study did not utilize ML-based downscaling (Reddy et al. 2023). In contrast, this research's RF and XGB models demonstrated superior performance metrics (e.g.  $R^2 > 0.9$  for key climate variables), reflecting significant advancements in model precision and adaptability. The projected increase in wet season discharge under SSP585 by approximately 10%, alongside a 40% reduction in dry season flows, is consistent with findings by Ligaray et al. (2015) in the nearby Chao Phraya Basin, where similar seasonal amplification was observed under high-emission scenarios (Ligaray et al. 2015).

Another notable contribution is the study's focus on uncertainty quantification. Like the approach by Chipman et al. (2006) in Bayesian ensemble learning, this study's BMA framework integrates multiple GCM outputs and accounts for spatial heterogeneity and localized climate dynamics (Chipman et al. 2006). The findings reveal critical insights into station-specific vulnerabilities, particularly under high-emission scenarios like SSP585, which previous studies often overlooked due to coarse spatial resolutions or simplified uncertainty analyses (Flato et al. 2014).

While this research excels in integrating advanced computational techniques and comprehensive datasets, some limitations persist. For instance, the performance of BMA varies across stations, with reduced accuracy in areas exhibiting complex hydrological dynamics. It is consistent with challenges noted by Hoeting et al. (1999), who highlighted the sensitivity of BMA to the quality of individual model inputs (Hoeting et al. 1999). Addressing these limitations may require incorporating additional predictors or refining model configurations to capture localized phenomena better.

One of the key sources of uncertainty in this study arises from the variability among the CMIP6 GCM projections, as differences in model structures, parameterizations, and climate sensitivity directly influence the future climate variables used in streamflow simulations. These uncertainties are further propagated into the hydrological modeling process, affecting the accuracy of streamflow estimates under future scenarios. Although BMA was applied to reduce individual model biases and enhance the reliability of the ensemble output, residual uncertainty remains, particularly regarding extreme events and localized hydrological responses. Additionally, limitations in the spatial resolution of GCMs, the quality of historical observations for model calibration, and assumptions inherent in emission scenarios (SSP245 and SSP585) contribute to the overall uncertainty. Future work should incorporate comprehensive uncertainty analyses, such as quantifying the spread of GCM outputs and applying probabilistic approaches to assess the confidence levels of projected streamflow changes.

This study contributes to the evolving field of hydrological modeling by bridging gaps in spatial resolution, model uncertainty, and predictive accuracy. Combining cutting-edge ML algorithms with ensemble modeling techniques provides actionable insights for regional water resource management and climate adaptation strategies. In addition, future studies are encouraged to incorporate comprehensive performance metrics such as the Combined Accuracy (CA) index, which integrates multiple error components to provide a more balanced evaluation of model performance (Adnan et al. 2019, 2024). These advancements align with and extend the methodological frameworks established in previous research, marking a significant step forward in climate change impact assessments.

## 5. Conclusion

In this study, the Multidimensional framework is built to robustly assess climate-driven streamflow variability in the TC-Basin by coupling CMIP6 GCMs, ML-based downscaling, and BMA. It overcomes the GCM limitations of spatial and temporal resolution and gets good predictive precision of both precipitation ( $R^2 = 0.90$ , RMSE = 2.26–2.37, MAE = 0.30–0.33 mm) and temperature (max  $R^2 = 0.92$ , RMSE = 0.78–1.43 °C). In particular, top performing GCMs in terms of precipitation and temperature were GFDL-ESM4 and CNRM CM6-1, respectively, while BMA ensembles improved at bias reduction to yield probabilistic precipitation forecasts with  $R^2 = 0.84$  (RMSE = 4.24–5.45 mm) and temperature projections with  $R^2 = 0.91$  (RMSE = 0.78–1.17 °C).

Under SSP585, hydrological simulations show pronounced shifts: increases in wet season (June–October) streamflow by 9–10% to enhance flood risk and decreases in dry season (March–April) flows by 40–43% to amplify water scarcity. On the other hand, cool, dry season (November–December) flows increase by 33–56%, potentially increasing irrigation availability. In terms of spatial distribution, maximum precipitation is 96,151.50 mm in annual precipitation under SSP585, and maximum temperatures are 33.66 °C in northern/central regions. The wind speed projections are moderately variable ( $R^2 = 0.88$ , RMSE = 1.80–3.03 m/s), and relative humidity is stable ( $R^2 > 0.89$ ). This study analysis offers actionable insights for adaptive water management in tropical monsoonal regions such as TC-Basin by disentangling, quantitatively, seasonal streamflow shifts under SSP585, where they contrast across wet, dry, and cool dry seasons with ( $\pm 40$ –56%) changes.

Despite this advancement, this study has limitations; there are no decomposition methods to remove noise from a varying climate dataset, and data scarcity in ungauged basins remains the limitation. Providing scalability of data sparse regions under the future work should have high-resolution remote sensing incorporated, to study out hybrid ML for

accuracy improvement. This framework highlights the importance of adaptive water management strategies to reduce climate risks under high-emission scenarios and a scalable methodology to run regional hydroclimatic assessments.

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## Disclosure statement

The authors declare that the research was conducted without any commercial or financial relationships that could be construed as a potential conflict of interest.

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